

No finite-rank operators do not force normality

A counterexample to Open Question 7.1 of arXiv:1009.3095

Automated mathematics run `fa_banach_001`, lane 2

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Status. Candidate counterexample, likely valid; prepared for human review.

1 Source question

Lord and Sukochev define a spectral triple $(\mathcal{A}, H, D)$ as a countably generated unital nondegenerate $*$ -algebra of bounded operators on a separable Hilbert space, together with a self-adjoint D such that $[D, a]$ is bounded for every $a \in \mathcal{A}$ and $\langle D \rangle^{-1} = (1 + D^2)^{-1/2}$ is compact. If n is the spectral dimension and $\langle D \rangle^{-n} \in \mathcal{M}_{1,\infty}$, they consider

$$\Phi_\omega(S) = \text{Tr}_\omega(S \langle D \rangle^{-n}), \quad S \in \mathcal{A}''.$$

Their Open Question 7.1 asks [1, Section 7.2, p. 28]:

If $\mathcal{A}''$ contains no finite rank operators is Φ_ω (the functional in (1.1)) normal?

As usual, “contains no finite rank operators” means that it contains no *nonzero* finite-rank operator.

for any trace τ on a two-sided ideal $\mathcal{J}$, and g a positive bounded Borel function such that $g(\Delta) \in \mathcal{J}$. Here Δ is the Laplacian (resp. noncommutative Laplacian) on the torus and τ is the Lebesgue integral on commutative torii (resp. the unique faithful normal finite trace on the type II_1 noncommutative torii).

If $\mathcal{N} \subset \mathcal{L}(H)$ is a von Neumann algebra, a positive compact operator T is $(\mathcal{N}, h)$ -dominated if, for some orthonormal basis $\{h_m\}_{m=1}^\infty$ of eigenvectors of T , there exists $h \in H$ such that $\|Ph_m\| \leq \|Ph\|$ for all projections $P \in \mathcal{N}$. If $\mathcal{J}$ is any two-sided ideal with positive singular trace f , and $0 < T \in \mathcal{J}$ is $(\mathcal{N}, h)$ -dominated, then $\text{f}(T)$ is a positive normal linear functional on $\mathcal{N}$.

7.2 List of open questions

Let $(\mathcal{A}, H, D)$ be a spectral triple (as defined in the first paragraph of Section 1) such that $\langle D \rangle^{-n} \in \mathcal{M}_{1,\infty}$ (as defined at (1.5)).

Open Question 7.1 (from Section 1). *If $\mathcal{A}''$ contains no finite rank operators is Φ_ω (the functional in (1.1)) normal?*

Open Question 7.2 (from Section 1). *What are necessary and sufficient relationships between $\langle D \rangle^{-n}$ and the $*$ -algebra $\mathcal{A}$ so that Φ_ω (the functional in (1.1)) is independent of ω (for various sets of generalised limits)?*

Open Question 7.3 (from Section 3). *Is $B \subset S_\infty(\ell^\infty)$ satisfying the Cesàro limit property (from [27, § 5]) a necessary and sufficient condition for equality of a set $K_\omega^\pm$ of positive measurable*

Figure 1: Open Question 7.1 in the source paper, printed page 28.

2 Idea of the proof

The hypothesis only excludes projections whose Hilbert-space ranges are finite-dimensional. It does not exclude an atomic von Neumann algebra whose minimal projections all have infinite multiplicity. We represent $\ell_\infty(\mathbb{N})$ with precisely that multiplicity and choose D with a two-dimensional lattice spectrum. The full operator $\langle D \rangle^{-2}$ has logarithmically divergent trace and Dixmier trace $1/2$, but the part over any fixed atom is trace class. Hence the associated functional vanishes on every atom and on every finite sum of atoms, while it is nonzero at the identity. Those finite sums increase strongly to the identity, which is exactly a failure of normality.

3 Construction and theorem

Let $K = L = \ell_2(\mathbb{N})$, with standard bases $(e_i)_{i \geq 1}$ and $(f_j)_{j \geq 1}$, and put $H = K \otimes L$. For $i \geq 1$, let

$$P_i = |e_i\rangle\langle e_i| \otimes I_L.$$

Define the unital $*$ -algebra

$$\mathcal{A} = \left\{ \lambda I_H + \sum_{i \in F} a_i P_i : \lambda, a_i \in \mathbb{C}, F \subset \mathbb{N} \text{ finite} \right\}.$$

It is generated by $I_H, P_1, P_2, \dots$. Define the positive self-adjoint operator D by

$$D(e_i \otimes f_j) = (i + j)e_i \otimes f_j$$

on its natural diagonal domain.

Theorem 1. *The triple $(\mathcal{A}, H, D)$ is a spectral triple of spectral dimension 2 with $\langle D \rangle^{-2} \in \mathcal{M}_{1,\infty}$. Its bicommutant contains no nonzero finite-rank operator. For every Dixmier trace Tr_ω in the conventions of the source paper, the positive tracial functional*

$$\Phi_\omega(S) = \text{Tr}_\omega(S \langle D \rangle^{-2}), \quad S \in \mathcal{A}'',$$

is not normal. In fact,

$$\Phi_\omega(I_H) = \frac{1}{2}, \quad \Phi_\omega(P_1 + \dots + P_r) = 0 \quad (r \geq 1).$$

Consequently, the answer to Open Question 7.1 is negative.

Proof. Write $H_i = P_i H = \mathbb{C}e_i \otimes L$. Since every element of $\mathcal{A}$ is scalar on each H_i , and since every P_i belongs to $\mathcal{A}$, its commutant is the algebra of bounded block-diagonal operators

$$\mathcal{A}' = \left\{ \bigoplus_{i \geq 1} X_i : X_i \in B(H_i), \sup_i \|X_i\| < \infty \right\}.$$

The commutant of this algebra consists exactly of independently chosen scalars on the blocks. Therefore

$$\mathcal{A}'' = \left\{ \sum_{i \geq 1} b_i P_i : (b_i) \in \ell_\infty(\mathbb{N}) \right\} \cong \ell_\infty(\mathbb{N}) \bar{\otimes} I_L. \quad (1)$$

If $0 \neq S = \sum_i b_i P_i$ belongs to $\mathcal{A}''$, choose i with $b_i \neq 0$. The restriction of S to the infinite-dimensional space H_i is $b_i I_{H_i}$, so S has infinite rank. Thus $\mathcal{A}''$ contains no nonzero finite-rank operator.

The algebra $\mathcal{A}$ is countably generated, unital and nondegenerate. Moreover D is diagonal in the same basis as every element of $\mathcal{A}$, so $[D, a] = 0$ for all $a \in \mathcal{A}$. Each eigenvalue $m \geq 2$ of D has multiplicity $m - 1$. It follows that $\langle D \rangle^{-1}$ is compact and, for $p > 0$,

$$\mathrm{Tr}(\langle D \rangle^{-p}) = \sum_{m=2}^{\infty} (m-1)(1+m^2)^{-p/2}.$$

The summand is asymptotic to m^{1-p} , so this series converges exactly when $p > 2$. Thus the spectral dimension is $n = 2$.

Set $T = \langle D \rangle^{-2}$. Its eigenvalue $(1+m^2)^{-1}$ has multiplicity $m-1$. At the end of the M th complete eigenspace shell there are

$$K_M = \sum_{m=2}^M (m-1) = \frac{M(M-1)}{2}$$

eigenvalues, and their sum is

$$S_M = \sum_{m=2}^M \frac{m-1}{1+m^2} = \log M + O(1). \quad (2)$$

Since $\log(1 + K_M) = 2 \log M + O(1)$, (2) gives

$$\frac{S_M}{\log(1 + K_M)} \longrightarrow \frac{1}{2}. \quad (3)$$

Between K_M and K_{M+1} the omitted part of one shell has total mass at most $M/(1+(M+1)^2) = O(M^{-1})$, so (3) holds along all partial sums of the decreasing eigenvalue sequence. In particular $T \in \mathcal{M}_{1,\infty}$, is Tauberian (Dixmier measurable), and the defining logarithmic mean has the ordinary limit $1/2$. Hence every normalized Dixmier trace under consideration satisfies

$$\mathrm{Tr}_\omega(T) = \frac{1}{2}. \quad (4)$$

For $r \geq 1$, put $F_r = P_1 + \cdots + P_r \in \mathcal{A}$. The positive operator $F_r T$ is trace class because

$$\mathrm{Tr}(F_r T) = \sum_{i=1}^r \sum_{j=1}^{\infty} \frac{1}{1+(i+j)^2} < \infty. \quad (5)$$

Every Dixmier trace vanishes on trace-class operators, and therefore $\Phi_\omega(F_r) = 0$. On the other hand, $F_r \uparrow I_H$ strongly, while (4) yields $\Phi_\omega(I_H) = 1/2$. A positive normal functional preserves such bounded increasing suprema. Thus Φ_ω is not normal.

Finally, (1) is abelian, so the positive functional Φ_ω is automatically tracial on $\mathcal{A}''$. This completes the verification of every hypothesis and the failure of the asserted conclusion. $\square$

4 Verification notes

The counterexample does not exploit a noncommutative-extension ambiguity: $\mathcal{A}''$ is abelian. Nor does it rely on a special generalized limit: the ordinary logarithmic eigenvalue mean exists, so (4) holds for every Dixmier trace in the source classes. Equation (5) is the decisive separation between Hilbert-space rank and Dixmier mass: F_r has infinite rank on H , as the hypothesis permits, but it sees only finitely many summable rows of T .

A small numerical script in `code/check_shell_asymptotics.py` checks the exact multiplicities, convergence of the ratios in (3), and stabilization of finite-row sums. It is a sanity check only; the proof is the comparison argument above.

5 Bounded novelty check and limitations

The local run indexes were searched for arXiv:1009.3095, the paper title, “no finite rank operators”, “Dixmier trace normal”, and close variants; no prior packet or attempt was found. Exact-phrases and close-variant web/arXiv searches were run for Open Question 7.1 and the normality conjecture. The OpenAlex record for the source (work W2101700424) listed 15 citing records through 9 August 2026, including duplicate arXiv/journal versions. Their titles were screened, and locally available full text was queried for arXiv:1201.3661, 1210.3397, 2103.02067, and 2412.00628. No later paper explicitly answering Question 7.1, and no occurrence of this construction, was found within those bounds. This is a bounded novelty audit, not a certification of priority.

The result answers only Open Question 7.1. It does not characterize normality and does not address the source paper’s other questions about measurability, generalized limits, or maximal symmetric subideals. It uses the deliberately broad spectral-triple definition printed in the source; a stronger geometric axiom system might exclude the example and would require a separately formulated question.

6 Human-review recommendation

Send to a specialist in noncommutative geometry or operator ideals. The two highest-value checks are: (i) the bicommutant computation (1), including the infinite-multiplicity representation, and (ii) the normalization $1/2$ in (3)–(4). If those pass, the monotone-convergence obstruction is immediate.

References

- [1] S. Lord and F. Sukochev, *Measure Theory in Noncommutative Spaces*, SIGMA 6 (2010), 072, 36 pp.; arXiv:1009.3095.